Applied Machine Learning Report

Vehicle Silhouette Classification, Capstone Project 3 (Applied Machine Learning)

Overview

This report presents a supervised, 4-class classification model built on the **Statlog (Vehicle Silhouettes)** dataset, available from the UCI Machine Learning Repository. The dataset contains 846 examples of four model-vehicle types (a bus, a van, and two car models), each described by 18 numeric shape features extracted from a 2D silhouette image. A logistic regression classifier, selected over a random forest ensemble based on cross-validation performance, was trained to predict vehicle type from these shape features; the full workflow is in the accompanying notebook (`modeling.ipynb`).

Dataset Description

The data were originally gathered at the Turing Institute (Glasgow) by photographing four Corgie scale-model vehicles, a double-decker bus, a Chevrolet van, a Saab 9000, and an Opel Manta 400, from many rotation angles, thresholding each image into a binary silhouette, and extracting 18 scale-independent shape measurements (e.g., compactness, circularity, aspect ratios, scaled moments, a hollows ratio) using the HIPS/BINATTS image-processing pipeline. After combining the dataset's nine raw chunk files into a single table, it contains 846 rows and 19 columns: the 18 numeric shape features plus the categorical target, `CLASS` (`bus`, `van`, `saab`, `opel`). The four classes are close to balanced (218 bus, 217 saab, 212 opel, 199 van), there are no missing values and no duplicate rows, making this a clean dataset with no missing-value imputation required.

Modeling Approach

The task was framed as supervised multi-class classification rather than clustering, since every silhouette already has a known vehicle-type label and the objective is to predict that label for new silhouettes. The 18 features differ widely in scale (small ratios near 0-1 alongside raw moment values in the hundreds), so a `StandardScaler` was applied before any scale-sensitive model; it was fit only on the training fold inside a scikit-learn pipeline to avoid leaking test-set statistics into preprocessing (Pedregosa et al., 2011). An 80/20 train/test split, stratified by class, preserved the original class balance in both subsets.

Two candidate models were trained and compared via 5-fold stratified cross-validation on the training set before any test-set evaluation: a scaled logistic regression as an interpretable linear baseline, and a random forest of 200 trees as an ensemble method that averages many trees trained on bootstrapped samples to reduce the variance any single decision tree would have (Breiman, 2001). Logistic regression achieved the higher mean cross-validation accuracy (0.781 ± 0.025 vs. 0.752 ± 0.011 for the random forest) and was selected as the final model. Accuracy, together with macro-averaged precision, recall, and F1, were used for evaluation rather than accuracy alone, since macro-averaging treats all four classes as equally important and would expose any single class the model struggles with even though the classes are fairly balanced here.

Results

On the held-out test set (170 rows), the final logistic regression model reached **82% accuracy**, with macro-averaged precision 0.82, recall 0.82, and F1 0.81. Performance was highly uneven across classes: the bus and van classes were classified almost perfectly (F1 = 0.96 for both), while the two car models were substantially harder to separate from each other (opel: precision 0.71, recall 0.52, F1 0.60; saab: precision 0.67, recall 0.80, F1 0.73).

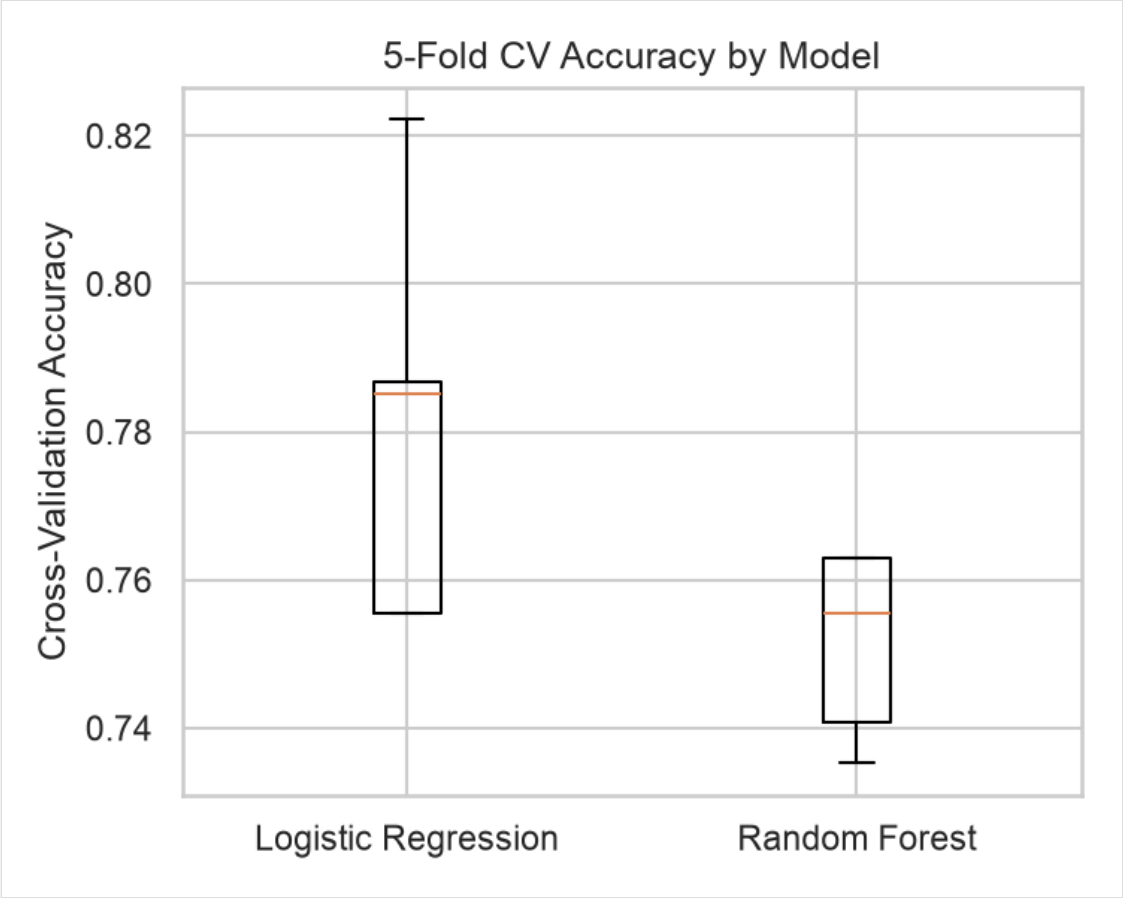


Figure 1. 5-fold cross-validation accuracy for the two candidate models on the training set. Logistic regression was both higher-accuracy and more consistent across folds.

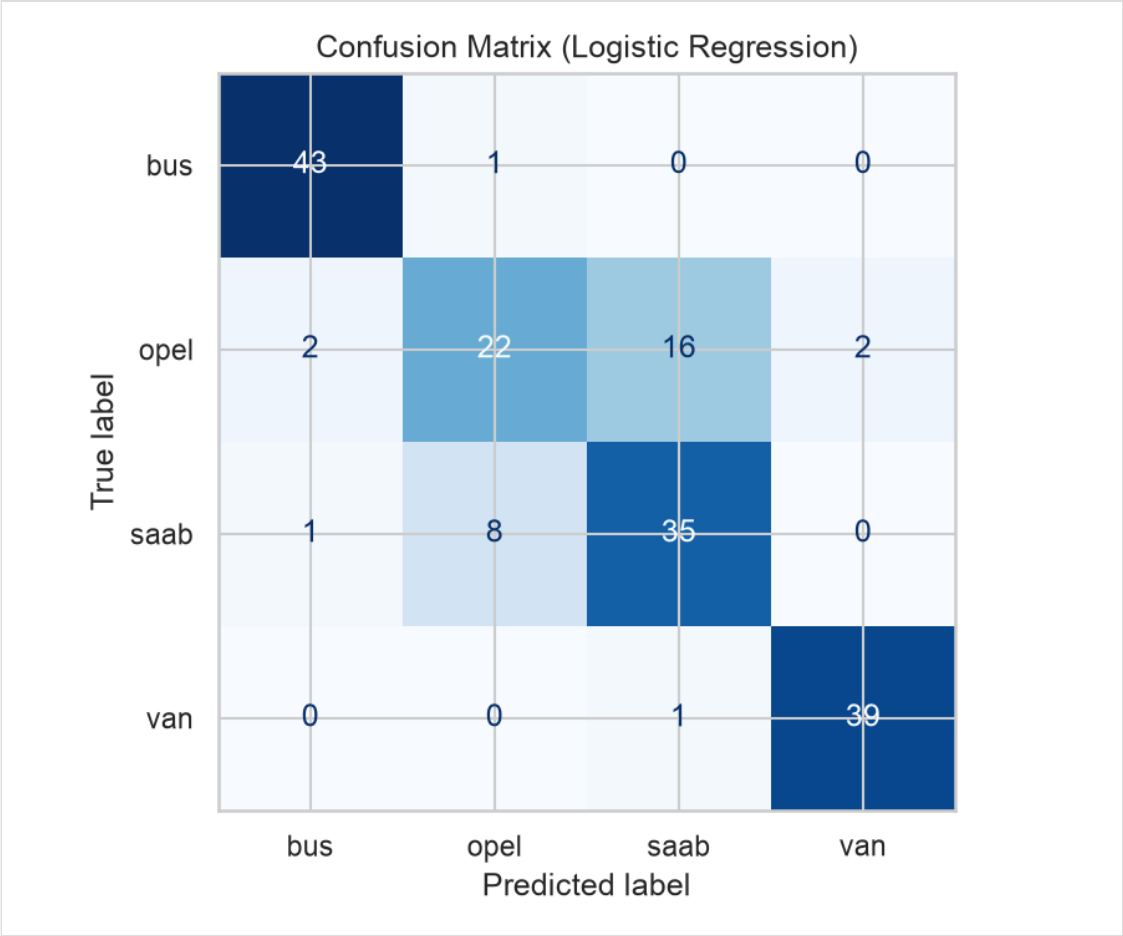


Figure 2. Confusion matrix on the held-out test set. Nearly all errors (16 + 8 of 170 rows) are opel/saab confusions; bus and van are essentially never confused with any other class.

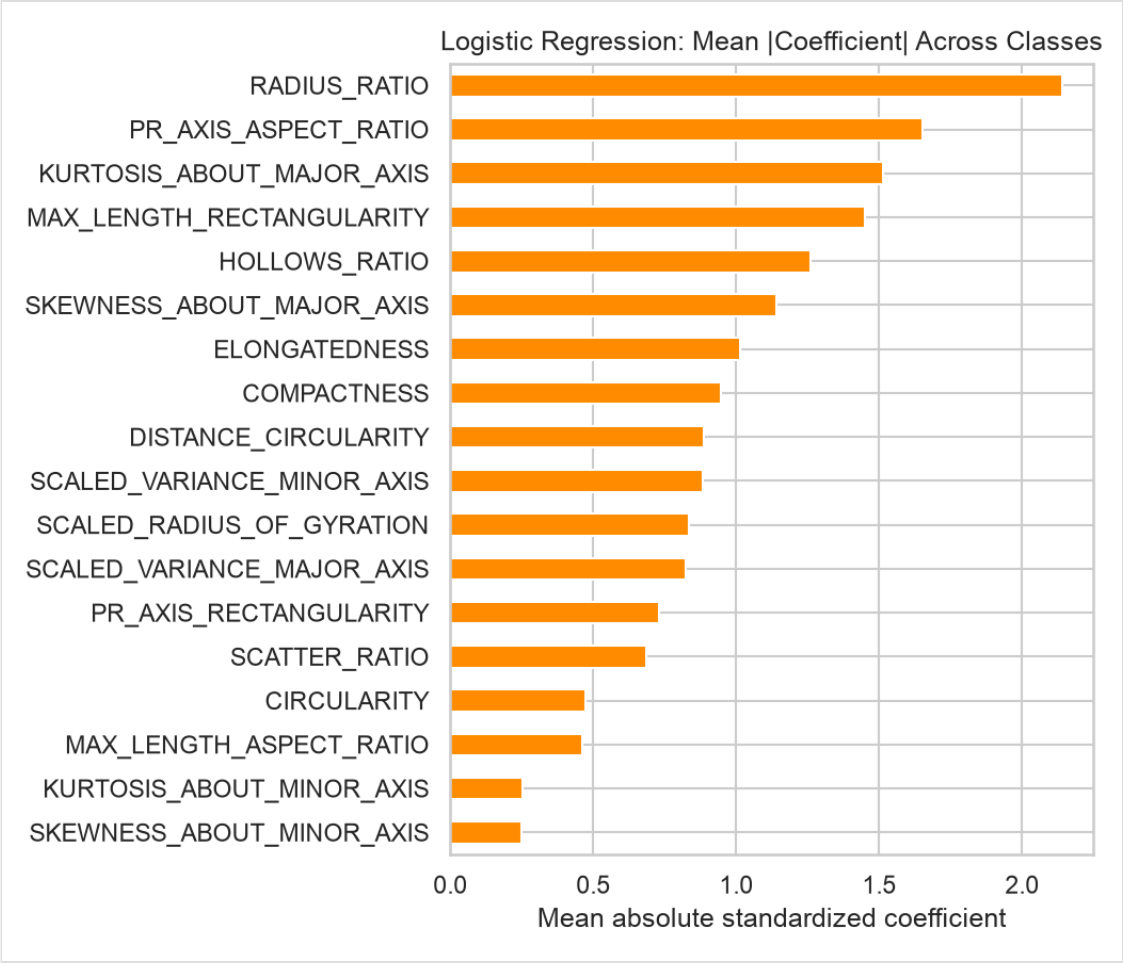


Figure 3. Mean absolute standardized logistic regression coefficient per feature, averaged across the model's one-vs-rest class comparisons. Radius ratio, principal-axis aspect ratio, and kurtosis about the major axis were the most influential shape features.

Interpretation for a Non-Technical Audience

In plain terms: given only a handful of numbers that describe the outline shape of a vehicle in a photo (how round it is, how long versus wide, how lopsided), the model correctly names the vehicle type about 4 times out of 5. It is excellent at the easy part of the problem, telling a bus or a van apart from either car almost never goes wrong, but it struggles with the hard part, telling the two car models (an Opel and a Saab) apart from each other, because two different sedan-shaped cars simply look more alike in outline than a van looks like a bus. This mirrors how a person might struggle to tell two similarly shaped sedans apart from a distant silhouette, while having no trouble telling a bus from a car.

Limitations and Potential Bias

Dataset limitation: several of the 18 shape features are strongly correlated with one another, since they are all derived mathematically from the same underlying silhouette outline (for example, the scatter ratio and the two rectangularity measures all capture overlapping notions of "how elongated or boxy" a shape is). This redundancy was managed by using L2-regularized logistic regression and by not treating any single feature's coefficient as fully independent evidence, but a model built on a feature set with less redundancy might reach a similar accuracy with a simpler, more directly interpretable set of inputs.

Potential source of bias: the dataset consists of exactly four specific scale-model vehicles photographed under controlled studio lighting and a fixed camera elevation, not a broad sample of real-world vehicles in varied conditions. A classifier trained on this data encodes the shape idiosyncrasies of these four particular model cars rather than vehicle types in general, so its errors (particularly opel/saab confusion) should be read as a property of these two specific models' silhouettes rather than a general claim about how distinguishable sedans are from one another. This is a narrow instance of a broader, well-documented risk in applied machine learning: a model's apparent accuracy can mask dataset-specific biases that would not hold up on a more representative sample, and practitioners are responsible for checking a model's training distribution before deploying it (Mehrabi et al., 2021). As a mitigation, any reuse of this model outside this specific four-vehicle dataset should be preceded by re-validating performance on a new, independently collected sample of silhouettes before trusting its predictions.

References

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